

Gowtham Sai Chimmana

AI / ML Engineer · LLMs & RAG Systems · NLP

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PROFESSIONAL SUMMARY

- **AI/ML Engineer building production-style LLM & RAG systems** — offline document Q&A, offline technical-doc summarization, and a live RAG resume-analyzer on Hugging Face Spaces, using LangChain, FAISS, PyTorch, TensorFlow, and FastAPI.
- **Full-stack across the AI product — model to UI**: 2 years shipping production software as a frontend engineer at an EdTech startup, now building the model layer (PyTorch/LangChain), backend (FastAPI), and frontend (React) end-to-end — able to own an AI feature from data to deployed interface without a hand-off.
- **CS graduate student at USF (GPA 3.50), Teaching Assistant for Operating Systems, contributing to faculty-led neural-symbolic AI research**. Seeking AI Engineer, ML Engineer, or Applied LLM roles. Authorized to work in the US (OPT).

TECHNICAL SKILLS

- **AI / ML**: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, BERT, RoBERTa, NLP, LLMs, RAG, MLflow
- **LLM & Agents**: LangChain, LlamaIndex, Pinecone, ChromaDB, Prompt Engineering, OpenAI API, Ollama, Gradio
- **Languages**: Python, JavaScript, HTML, CSS, SQL, Bash, Java, C++
- **Frontend & UX**: React.js, REST API integration, Figma (fundamentals), UX Design Principles, Responsive Design
- **Backend & MLOps**: FastAPI, Docker, Git, CI/CD, Linux, PostgreSQL, MongoDB, Microsoft Azure, PySpark
- **Frameworks**: OpenCV, PyMuPDF, NumPy, Pandas
- **Certifications**: Microsoft Azure AI-900 · Oracle Java SE 8 · ServiceNow Certified System Administrator

AI / ML PROJECTS

LLM-Powered Technical Document Validation Assistant — *Python · LangChain · Llama 3 · Gradio · Ollama*

- Teams reviewing long technical specs and compliance documents often can't use cloud AI tools due to data confidentiality — built a fully offline assistant that delivers AI summarization with zero data leaving the organization's infrastructure and zero cloud compute cost.
- Eliminated cloud API costs and data-egress risk entirely by architecting a fully offline map-reduce summarization pipeline — PyMuPDF semantic chunking feeding a locally-hosted Llama 3 (via Ollama) through LangChain.
- Cut summary turnaround to under 8 seconds and eliminated context-overflow failures on documents up to 500+ pages, reaching 91% user-rated summary quality across engineering specs and research papers.

AI-Powered Offline Document Q&A Assistant — *Python · PyMuPDF · NLP · Keyword Scoring*

- Built for scenarios like a field engineer needing an instant answer from a 300-page manual without GPU resources or internet access — engineered a lightweight retrieval system that runs on any machine, with no cloud inference costs.
- Engineered a GPU-free, internet-free document Q&A system using keyword scoring, phrase matching, and query expansion to answer natural language queries in under 2 seconds.
- Achieved 88% answer relevance on a manual benchmark, cutting information retrieval time by 65% versus manual PDF scanning.

AI Career Coach — RAG Resume Analyzer — *Python · LangChain · FAISS · FastAPI · Gradio · Hugging Face Spaces* | [Live Demo](#)

- Job seekers often can't tell which specific skills are missing between their resume and a target role — built a tool that pinpoints the gap and generates a concrete learning plan instead of generic advice.
- Built an end-to-end RAG pipeline embedding resume and JD text with sentence-transformers, storing vectors in FAISS, and retrieving semantically relevant skills for gap analysis.
- Designed a LangChain prompt chain that compares candidate skills against job requirements and generates a personalized learning roadmap using Llama 3 — deployed live and publicly accessible on Hugging Face Spaces.

AI/ML Visual Defect Inspection System — *Python · TensorFlow · Keras · FastAPI · TFLite*

- Manual visual quality inspection on manufacturing lines is slow, inconsistent across inspectors, and prone to fatigue-driven errors — built an automated first-pass classifier so human reviewers only need to check ambiguous cases instead of every unit.

- Trained a CNN defect-detection model on casting product images, reaching 96.4% classification accuracy across PASS/FAIL/REVIEW categories on a 2,000+ image test set.
- Compressed the model to TensorFlow Lite, cutting size 91.7% (127.87 MB → 10.66 MB) while retaining accuracy within 1.2%, enabling sub-100ms real-time CPU inference.
- Exposed predictions via a FastAPI REST endpoint with a monitoring dashboard tracking confidence scores and inspection rates.

WORK EXPERIENCE

Frontend Engineer | Phoenix Global (EdTech startup serving IIT/IIM tutors)

Jun 2022 – May 2024 · Started as intern; offer extended to 2 years as the team's sole frontend engineer, based on performance · Remote

- Owned UI/UX direction for the platform as design lead — ran brainstorming sessions with the design team and translated concepts into a clean, intuitive interface used daily by thousands of students and tutors.
- Built and shipped the company's core tutoring website from the ground up — the primary product experience used by students and tutors — using JavaScript, HTML, and CSS.
- Designed and built real-time dashboards for tutor availability, live course data, and student progress by connecting the frontend to backend learning management systems via REST APIs.
- Built Python and JavaScript automation for data validation and formatting between the LMS backend and frontend dashboards, reducing manual QA time and data-entry errors across course, tutor, and student progress data.
- Improved page load performance and mobile responsiveness across the platform by optimizing frontend rendering and reducing unnecessary re-renders — contributing to measurable gains in user engagement and session duration.

Graduate Teaching Assistant — Operating Systems | University of South Florida

Nov 2024 – May 2026 · Tampa, FL

- Supported Prof. Valentina Korzhova's research on neural-symbolic AI for natural language understanding — contributing to experiments in automatic text parsing, semantic reasoning, and linguistic semantics that advanced 2 active faculty publications.
- Instructed and mentored 60+ graduate students per semester in Operating Systems — covering process scheduling, memory management, concurrency, deadlocks, and synchronization — achieving 92%+ course satisfaction.
- Designed 12+ coding assignments and lab exercises on CPU scheduling, virtual memory, and synchronization primitives aligned with ACM CS curriculum standards.
- Built Python automation scripts for grading OS simulation projects, cutting turnaround time by 35% and resolving 100+ student queries per semester through structured office hours.

EDUCATION

Master of Science — Computer Science | University of South Florida

Aug 2024 – May 2026 · Tampa, FL · GPA: 3.50 / 4.0

Coursework: Machine Learning, NLP, Distributed Systems, Advanced Algorithms

B.Tech — Computer Science & Engineering | KL Deemed to be University

Aug 2020 – May 2024 · India · GPA: 8.57 / 10.0

Coursework: Data Structures, Operating Systems, AI, Database Management